

S V V Sai Sri Vallabh Pataneni

Master's Thesis: Hydrogen Energy-Systems Analysis & Optimisation

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SUMMARY

An Automotive Engineering Master's student at RWTH Aachen with a strong focus on energy-systems modelling and optimisation. At Robert Bosch I built and optimised MATLAB/Simulink models of fuel-cell powertrain strategies using Genetic Algorithms, benchmarked against dynamic programming and ECMS, and I have built a Well-to-Wheel and total-cost-of-ownership model comparing powertrains over their life cycle. I am seeking the Master's thesis at the FZ Jülich Institute of Climate and Energy Systems to analyse the import potential of green hydrogen derivatives for the German energy market.

KEY SKILLS

Modelling & optimisation: MATLAB/Simulink · Genetic Algorithms · Gradient Descent · DP/ECMS benchmarking · scenario analysis

Energy & techno-economics: Well-to-Wheel & TCO modelling · lifecycle analysis · fuel-cell systems · energy management

Programming & data: Python (NumPy, Pandas, scikit-learn) · Power BI · C/C++

Simulation: ANSYS (Fluent) · Star-CCM+ · Dymola

PROFESSIONAL EXPERIENCE

Operating Strategy / System Simulation Intern, Robert Bosch GmbH · 09/2025 – 12/2025

Schwieberdingen, Germany

- **Operating-Strategy Modelling:** Developed and validated an iterative Simulink model of a rule-based power-split operating strategy in the (FC)EVsim PHEV module, correlating outputs against reference Excel and measurement data for high model fidelity.
- **Multi-Objective Optimisation:** Executed multi-variable parametric optimisation using Genetic Algorithms and Gradient Descent to generate control MAPs for charge sustenance and minimal fuel consumption, benchmarked against DP and ECMS.
- **Analysis & Documentation:** Built post-processing and KPI analysis (power distribution, SOC window, efficiency, degradation) and co-authored a peer-reviewed paper on the rule-based FCEV control strategy.

Werkstudent, Fuel-Cell Systems (Aircraft), DLR Innovation Center for Small Aircraft Technologies (INK) · 01/2026 – Present

Würselen (Aachen), Germany

- **Fuel-Cell System Development:** Developing the ~150 kW fuel-cell system (the engine) for D-LIGHT, DLR's hydrogen-powered 9-seater feeder aircraft.
- **Thermal-Management Digital Twin:** Built a MATLAB/Simulink digital twin of the complete thermal-management system to derive design specifications (volumetric flow, coolant temperature range, coolant pressure), driving procurement of the coolant pump and heat exchanger.

Mechanical / Suspension Team Lead & Pit Crew, Ecogenium e.V (RWTH student team) · 06/2024 – Present

Aachen, Germany

- **Energy-Management Strategy:** Architected and deployed energy-management and fuel-cell temperature-control strategies in Python/Simulink (MIL-validated), boosting on-track fuel efficiency by ~80% and securing a runner-up finish at Shell Eco-Marathon 2025.

EDUCATION

M.Sc. Automotive Engineering, RWTH Aachen University · 09/2023 – 06/2026

Aachen, Germany · GPA 1.7 (1.2 in core automotive courses) · Fuel-Cell System Technology, Alternative/
Mobile Propulsion, Simulation Sciences, FEM, CFD

B.Tech Mechanical Engineering, JNTU Hyderabad · 06/2019 – 04/2023

Hyderabad, India · GPA 1.3 (German scale)

LANGUAGES

English (Fluent, C1) · German (Intermediate, B1) · Telugu / Hindi (Native)

SELECTED PROJECTS

- **Well-to-Wheel & TCO analysis, EV vs ICE:** built a lifecycle sustainability and total-cost-of-ownership model comparing powertrains, the same techno-economic framing as a hydrogen-import assessment.
- **Vehicle sensor-data analysis (IKA-RWTH):** Python pipelines for large datasets, KPI evaluation and Power BI dashboards.